Department of Mechanical Engineering

B.Tech Project Presentation on

VOLLEY: Design and Computational Feasibility Assessment of a Commanded Electromagnetic CubeSat Deployment Mechanism



Presented By

Adityavardhan Mishra · Pratham Chawla

PRN 23070125054 · PRN 23070125029

Guide: Vikas Gulia

Review candidate: 21 October 2026

SYMBIOSIS INSTITUTE OF TECHNOLOGY PUNE

Presentation route

OUTLINE

01 Introduction

02 Literature review

03 Research gap

04 Objectives

05 Research methodology

06 Timeline

07 Work progress

08 Results

09 Conclusion

10 Future work

11 References

Which missions might value orbital choice?

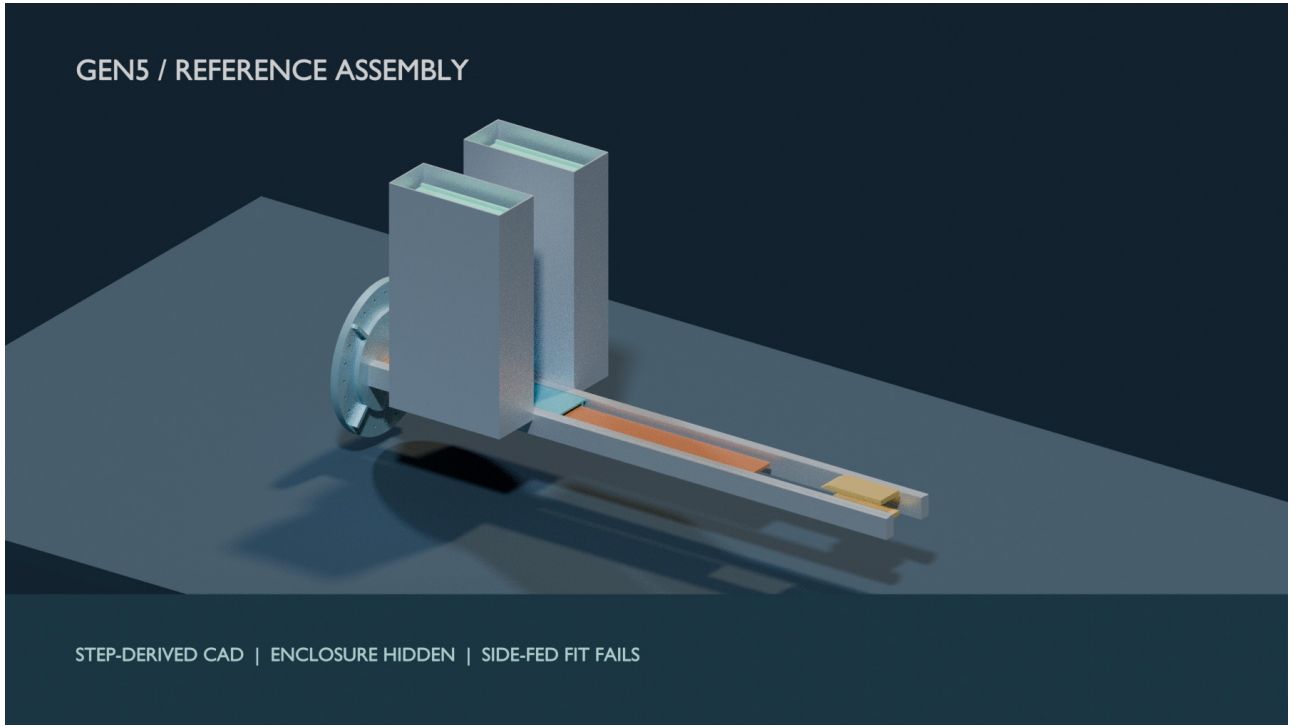
INTRODUCTION

Repeat fleets

Configurable satellite fleets exist; demand for VOLLEY is unconfirmed.

Delivery buyer

A carrier or fleet operator might seek several release states from one host.



STEP-derived Gen5 reference; side-fed placement fails its fit check.

Test on one matched mission against springs, host manoeuvres, transport and onboard propulsion.

NASA 2026 Small Spacecraft SoA; local market audit; STEP-derived Gen5 CAD

Existing methods solve different parts of the problem

LITERATURE REVIEW

Approach	Strength	Limit for this thesis
Spring deployer	Flight heritage; compact; simple	Limited tailored release speed
Spring release / drag	Relative state or drag can create phasing	Timing alone, with no relative state change, cannot
Orbital transfer vehicle	Larger orbit changes	Greater bus, mass and integration burden
Electromagnetic research	Commandable acceleration studied	Armature, high-g or maturity constraints vary

JAXA; Foster et al. 2018; Feng et al. 2025; Zhao et al. 2022/2025

A narrower, testable contribution

RESEARCH GAP

Commanded release for an unmodified CubeSat at moderate acceleration, with the full system counted.

Contribution examined

Gen5 reusable-sled linear motor, 3U payload, magazine, pulse power, arrest and orbital utility.

Boundary

Electromagnetic CubeSat deployment exists in prior work. Payload qualification and physical performance remain unproved.

The analytical research objectives have outcomes

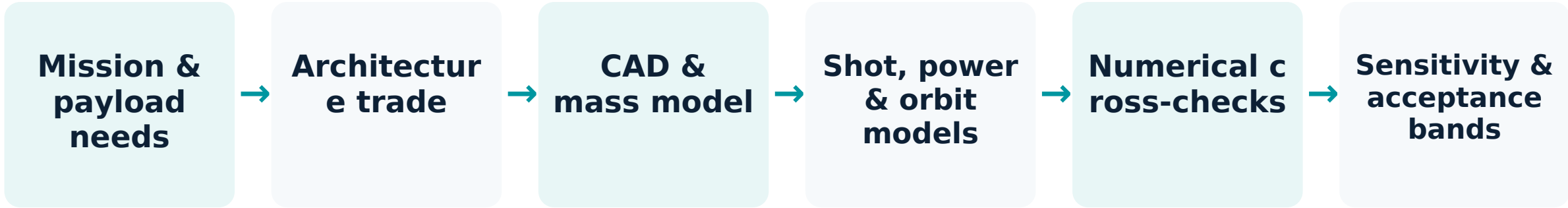
OBJECTIVES

Research objective	Final thesis outcome
Design a commandable 3U system	Reference CAD and subsystem concepts documented; feed remains open
Calculate shot and control performance	Historical model completed; finite force screen challenges its rating
Evaluate orbital utility	Conditional orbit results; real lifetime unvalidated
Count full installed burden	Mass rollup completed; 3U parity criterion failed
Test credibility of key claims	Numerical cross-checks and limits recorded

Thesis manuscript, BASELINE.md and PROVENANCE.md; presentation synthesis

The analysis chain from mission to decision

RESEARCH METHODOLOGY



Models were challenged and revised; hardware performance remains outside this thesis's evidence.

Numerical checks changed the review baseline

RESEARCH METHODOLOGY

Question	Method / cross-check	Evidence limit
Field and thrust	Analytic; 2-D FEM; independent 3-D numerical force	Shared ideal inputs; no motor measurement
Shot and power	Dynamics; Monte Carlo; ngspice	Assumed components and interfaces
Structure and flow	CAD; CalculiX; OpenFOAM	No payload qualification
Orbital change	Orbit model; Cowell / GMAT cases	Lifetime depends on atmosphere

Project evolution: documented decisions and analysis

TIMELINE

2023	**Hosted concept**	Free-flyer concept reframed around a host platform
Mid-2025*	**LSM direction**	Coilgun study gives way to a linear synchronous motor
2025–26*	**CAD generations**	Geometry and mass models mature through Gen5
Jul–Sep 2026	**Gen5 evidence**	Cross-checks, defect corrections and thesis record
Oct 2026	**Final review**	Gen5 results, failed criteria and thesis record prepared

* Approximate historical milestone; not a documented semester deadline.

VOLLEY-thesis/appendix/HISTORY.md; thesis README; PLAN-R2

2026 findings changed the review baseline

TIMELINE

Jul	CAD-derived sled mass replaced an optimistic parametric estimate
Aug	Depth-resolved thrust calculation lowered the historical model speed
Aug	Enclosure buildup exposed the full 3U mass penalty
2026	Independent orbit checks rejected a solar-invariance claim
Final	Results and failures retained in the submitted thesis record

HISTORY.md; PROVENANCE.md; thesis manuscript, Limitations

The current review package is assembled and traceable

WORK PROGRESS



FreeCAD-linked STEP envelope; internal interference is concealed.

Analysis delivered

Shot, circuit, structural, thermal, orbit and alternative-case models.

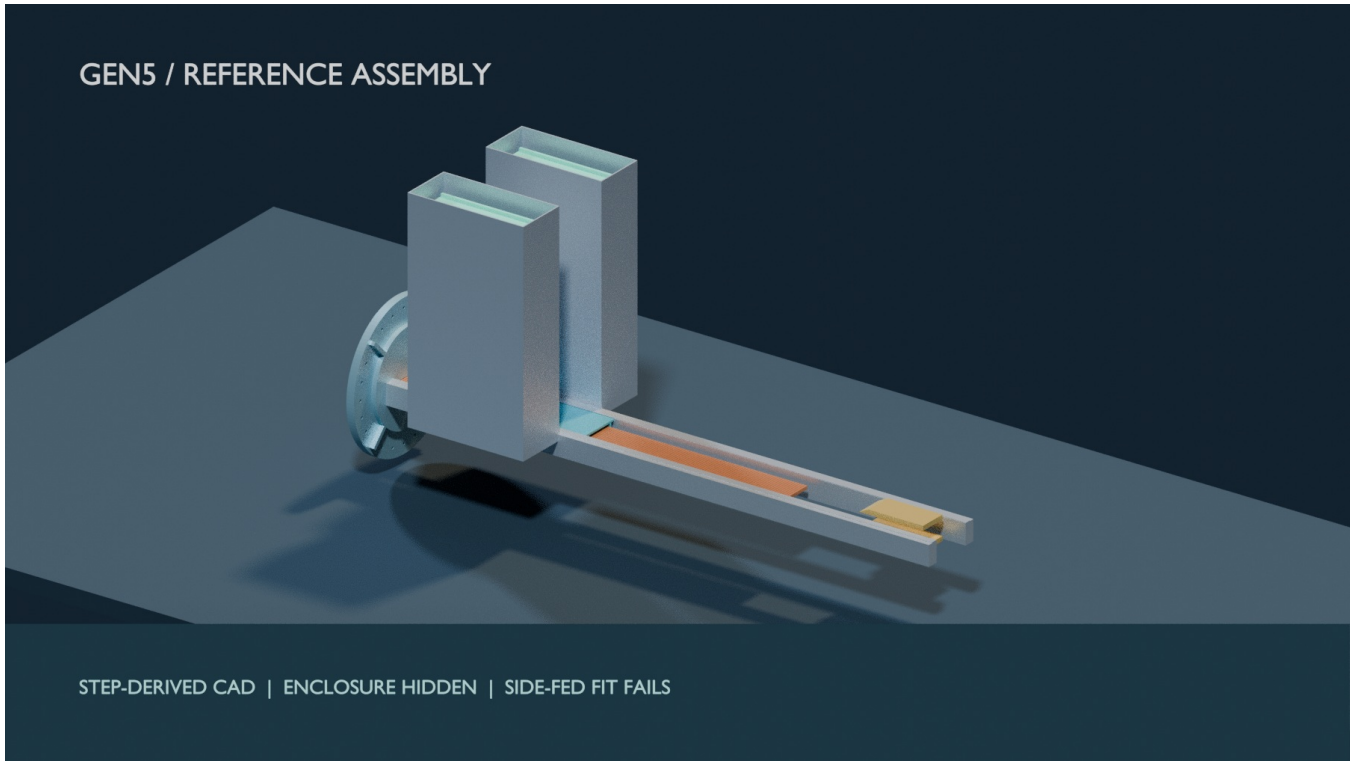
Evidence delivered

Manuscript, STEP, scripts, numerical checks and defect register.

Fixed computational study with failed design gates; no physical qualification claim.

Gen5: fixed reference geometry, feeder fit still open

RESULTS

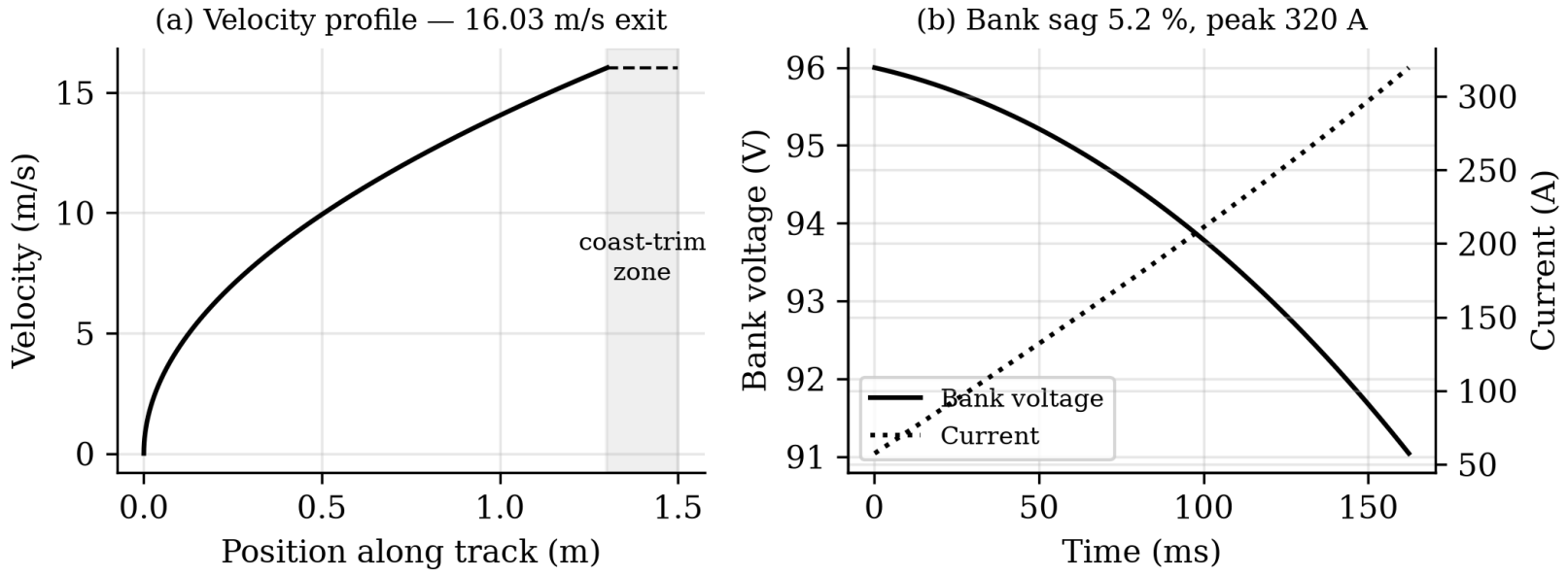


- 1.3 m powered stroke; 1.5 m release station
- Reusable 9.45 kg magnet sled
- Two cassettes; twelve 3U satellites
- FreeCAD STEP assembly: side-fed placement clashes

Enclosure hidden for visibility; no fabricated article.

Historical periodic-model shot: challenged by finite geometry

RESULTS



16.0 m/s
exit speed

10.07 g
acceleration

2.782 kJ
gross draw

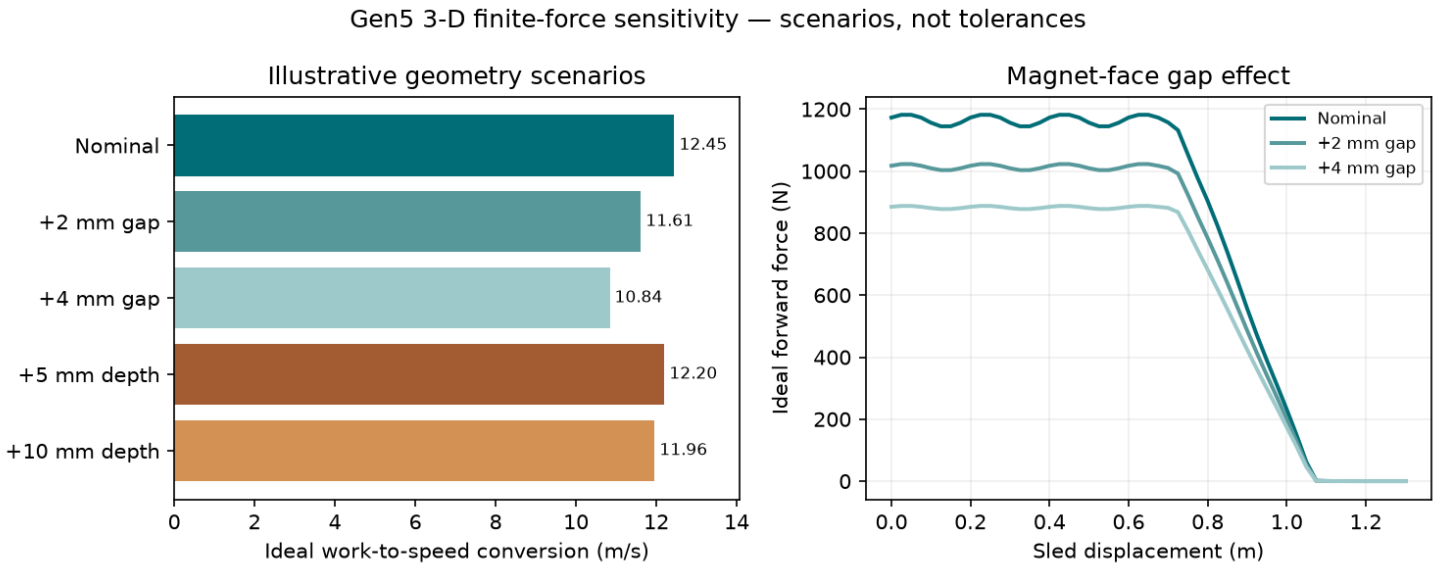
18.8%
net efficiency

P118 challenges 16.0 m/s; P117 reconciles 124.5 J to assumed model terms.

Thesis BASELINE.md; F01_shot.png; P117 energy audit

Independent 3-D check confirms ideal finite-force work

RESULTS



1.042 kJ

Ideal 3-D work: independent surface-charge formulation agrees with P118 under shared inputs.

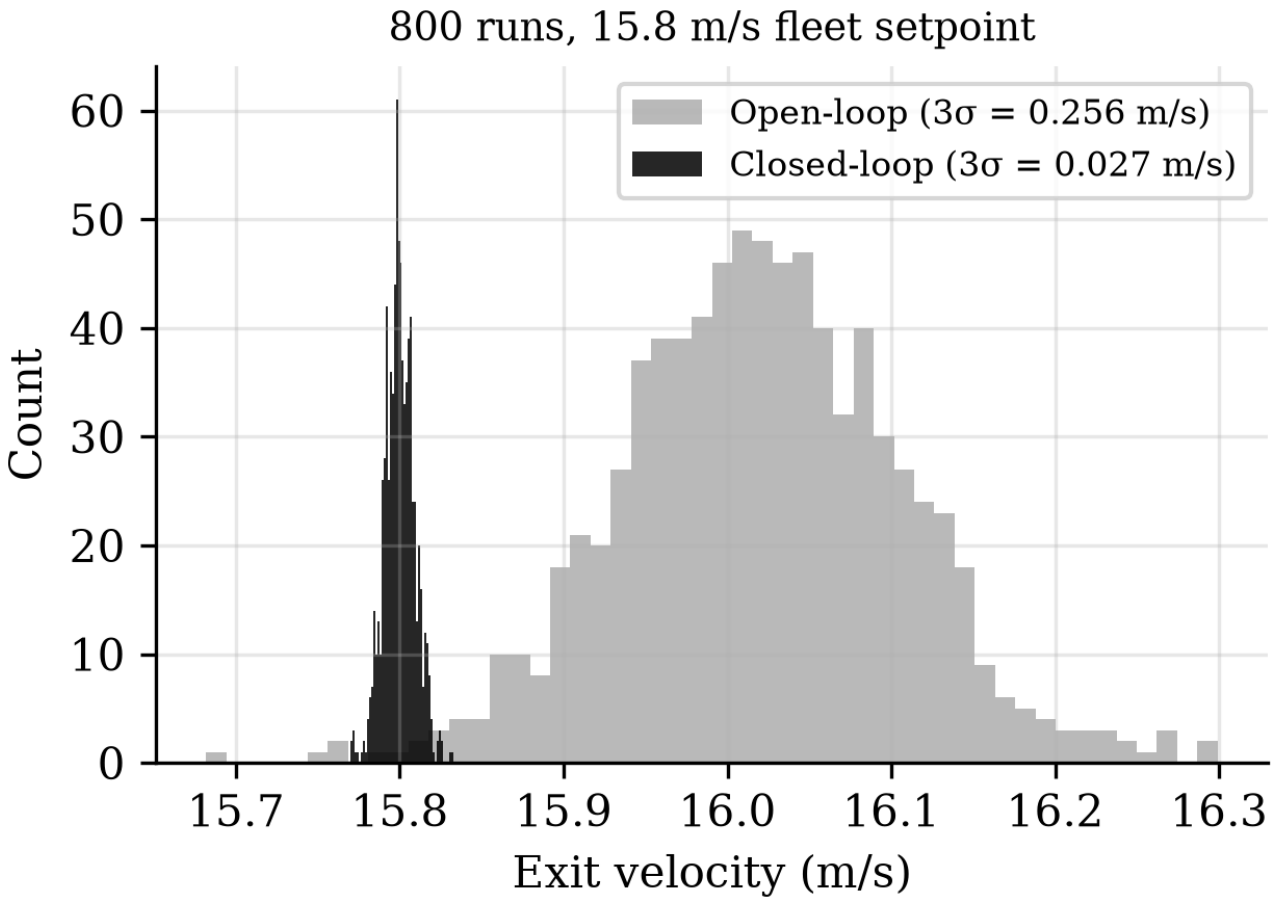
−13% work

Illustrative 12-to-14 mm magnet gap change; not a measured tolerance or motor rating.

No measured force, selected inverter or demonstrated release speed exists.

Modelled command precision is narrow; it is unmeasured

RESULTS



0.0274 m/s

3σ exit-velocity dispersion at a 15.8 m/s fleet setpoint.

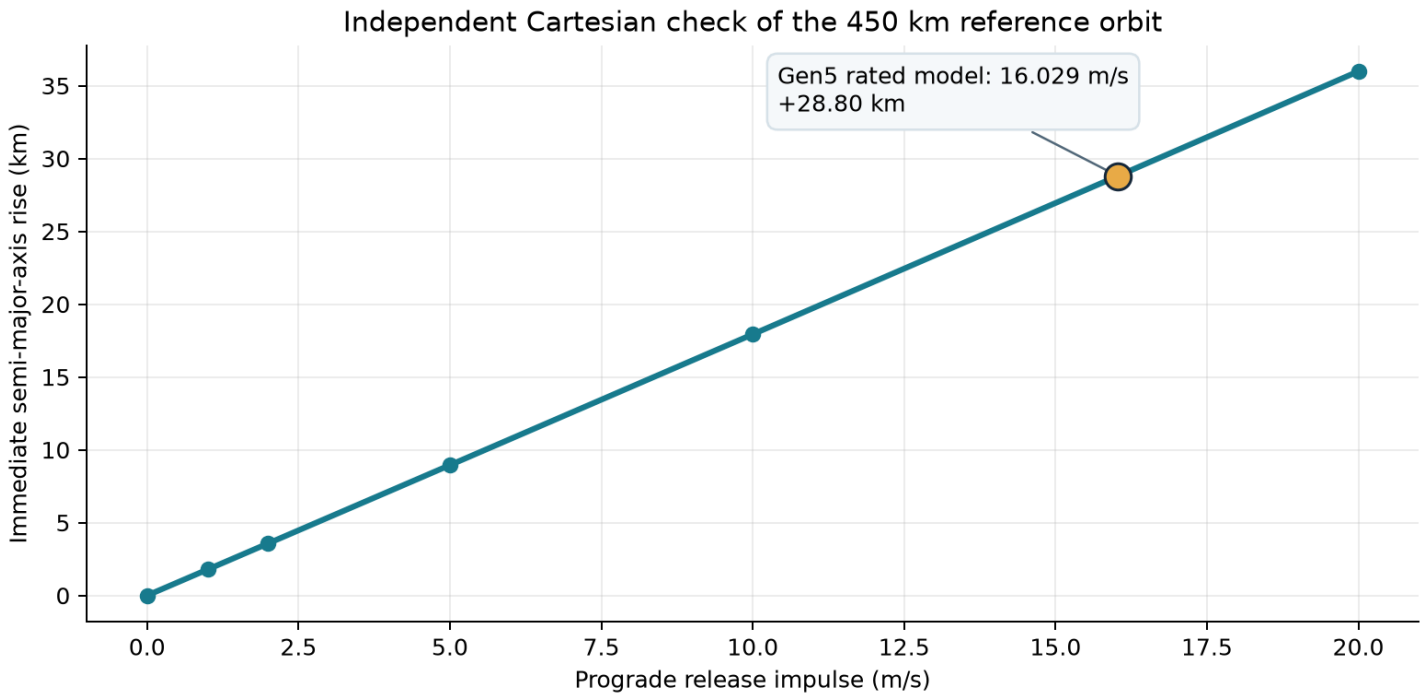
Evidence limit

Monte Carlo under modelled sensing and plant tolerances; no measured dispersion.

Thesis BASELINE.md; PROVENANCE.md; F03_mc.png

Historical orbit input has a separate geometry check

RESULTS



+28.8008 km

Immediate two-body axis rise for an assumed historical 16.029 m/s input.

1.602 model

P134 checks the shared static atmosphere; actual lifetime and achieved release remain unvalidated.

Persistent phasing requires a relative state change; this result concerns orbital energy.

Gen5 fails its 3U mass comparison

RESULTS

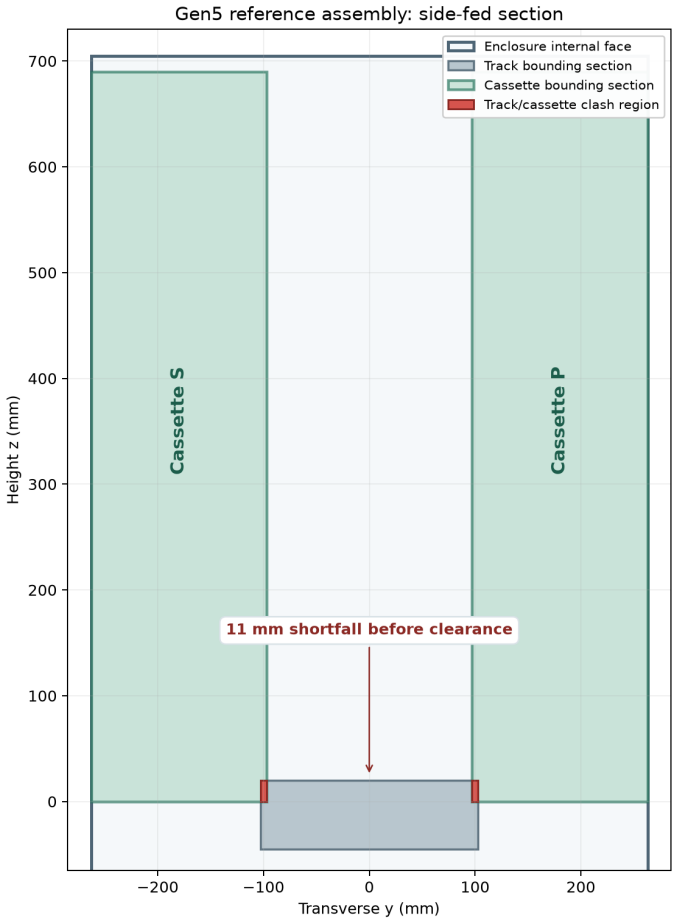
Modeled dry mass, 12 slots — **126.6 kg**



1.76× heavier per 3U payload; the preset 15% parity band fails.

The current side-fed assembly fails a geometric fit check

RESULTS



11 mm short

537 mm of track and cassette width inside 526 mm of usable enclosure width.

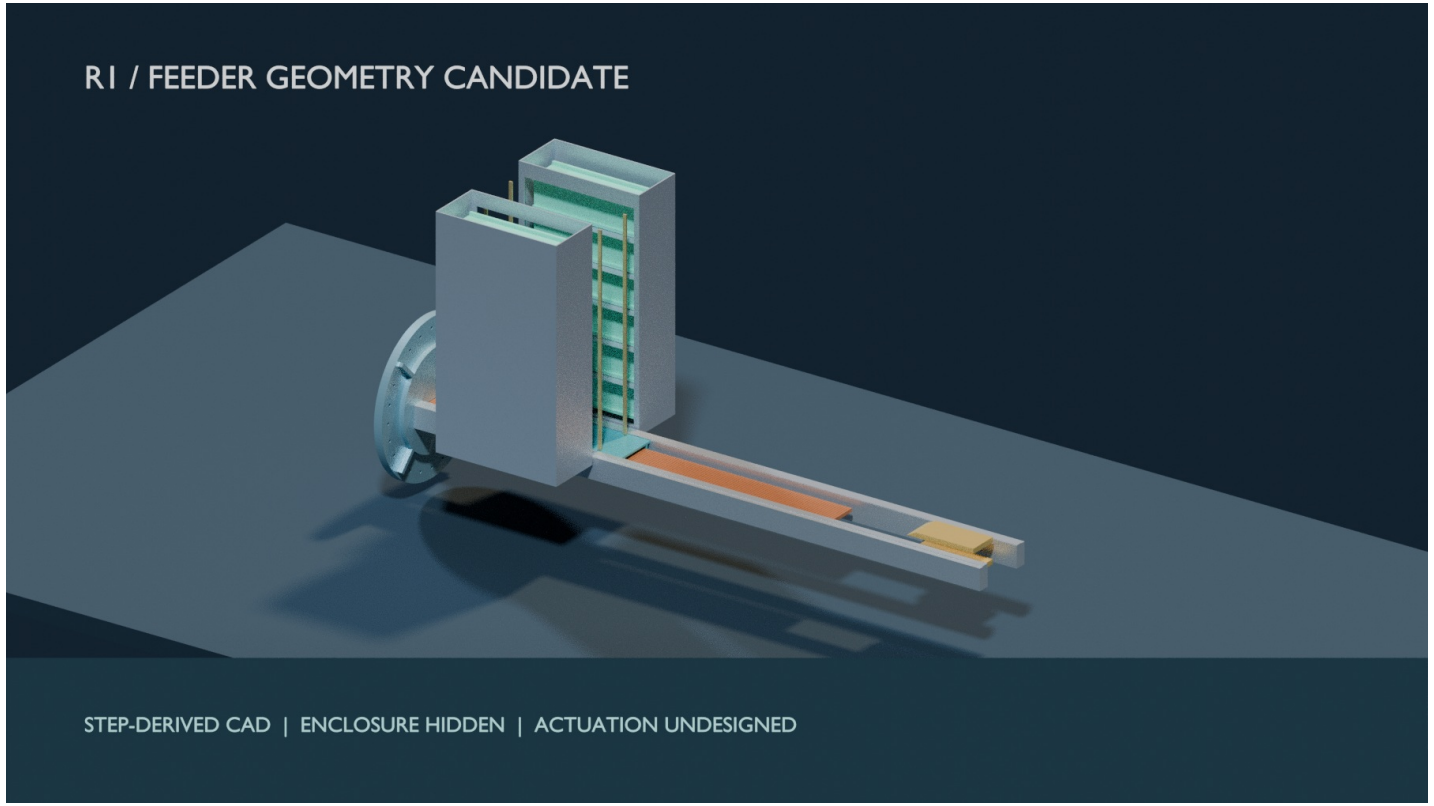
32,915 mm³

Cassette/track clash per side. Full audit: 17 intersecting pairs, including sled/track and sled/stator.

Reference placement only: a new feeder or enclosure would need a new configuration and rerun.

R1 clears cassette width, but still has assembly clashes

RESULTS



12 / 12

Selected scripted 3U paths clear; a whole-assembly audit still finds five intersecting solid pairs.

570 mm

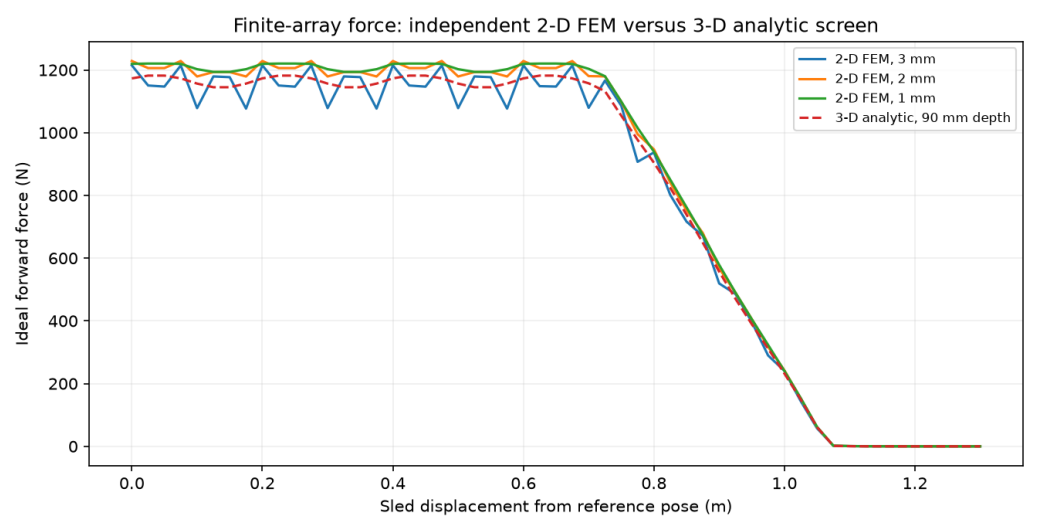
Wider enclosure; 0.5° assumed alignment stack fails its 5 mm gap; lift and retention still open.

Separate R1 candidate; no change to evaluated Gen5 shot, mass or host-fit claims.

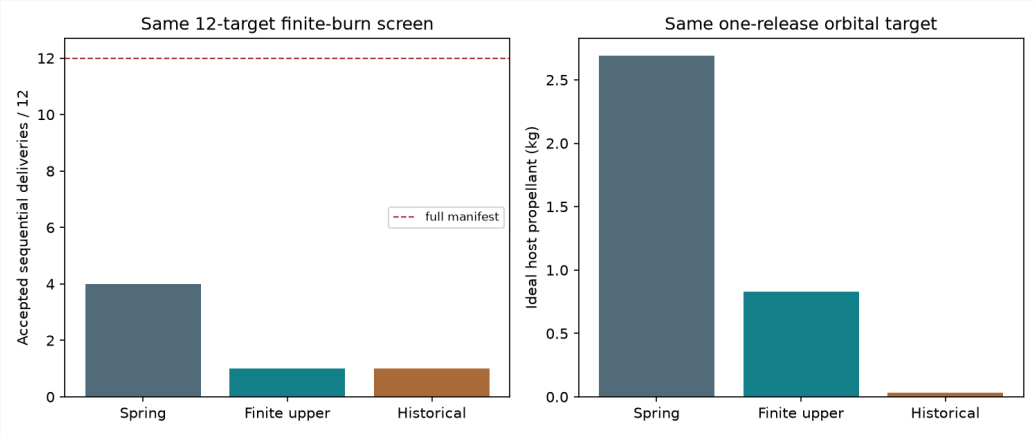
Computed results: from force to mission decision

RESULTS

1. Finite-force geometry screen



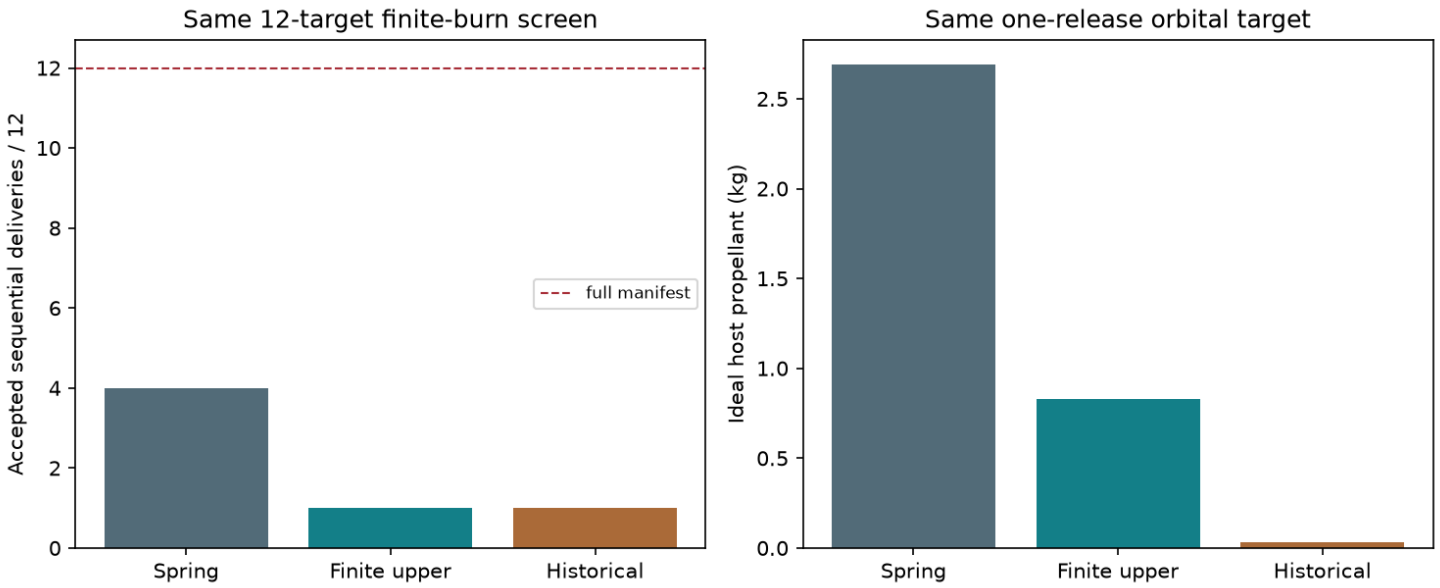
2. Matched mission screen



P125: single-string source power fails at the older commercial ESR bound; no twelve-shot case closes.

Matched reference mission finds no twelve-shot closure

RESULTS



+52.7 kg

One-event device plus resized fuel: Gen5 exceeds spring despite an ideal 1.87 kg fuel saving (P123).

4 / 1 / 1

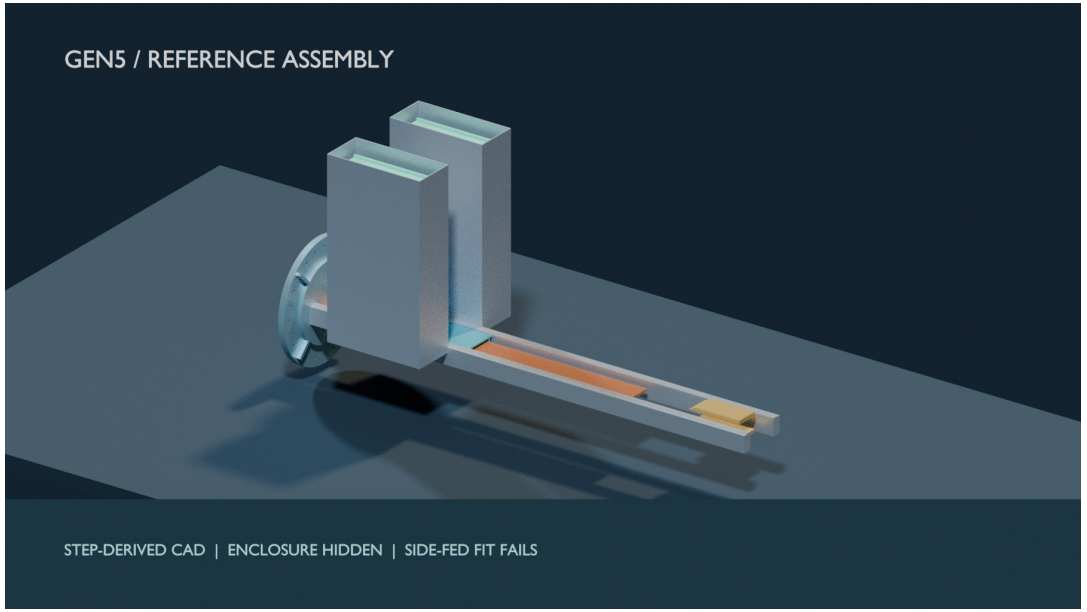
Accepted twelve-shot prefixes: spring, finite Gen5, historical Gen5.

Assumed host and devices; optimizer result is not an infeasibility proof or product advantage.

This Gen5 configuration does not pass selection

CONCLUSION

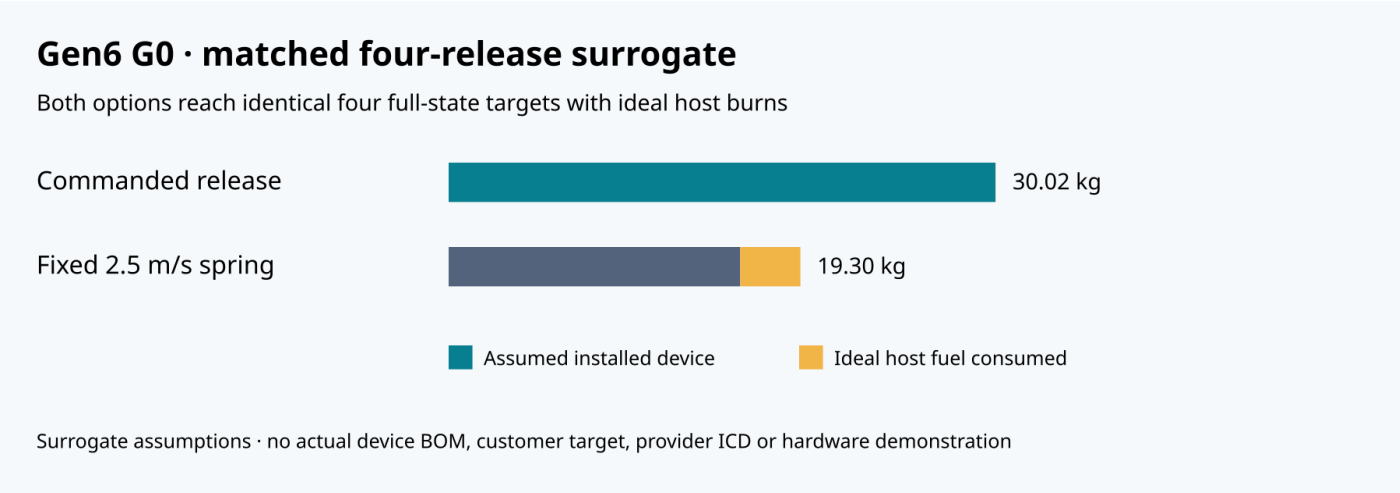
- Historical 16.0 m/s is challenged; ideal 12.448 m/s lacks a sourceable single-string bank.
- The modeled system is 76% heavier per 3U customer and the evaluated side-fed CAD clashes.
- R1 retains sled/track and sled/stator clashes; its assumed 0.5° gap stack fails.
- No matched twelve-shot reference closes; Gen5 is a documented negative design study.



STEP reference, enclosure hidden; fit fails. No physical test.

Gen6 has a testable mission, but no selected article

FUTURE WORK



Matched mission

Two 1U + two 3U targets: both release choices reach ideal states. Assumed device + fuel favors springs by 10.72 kg.

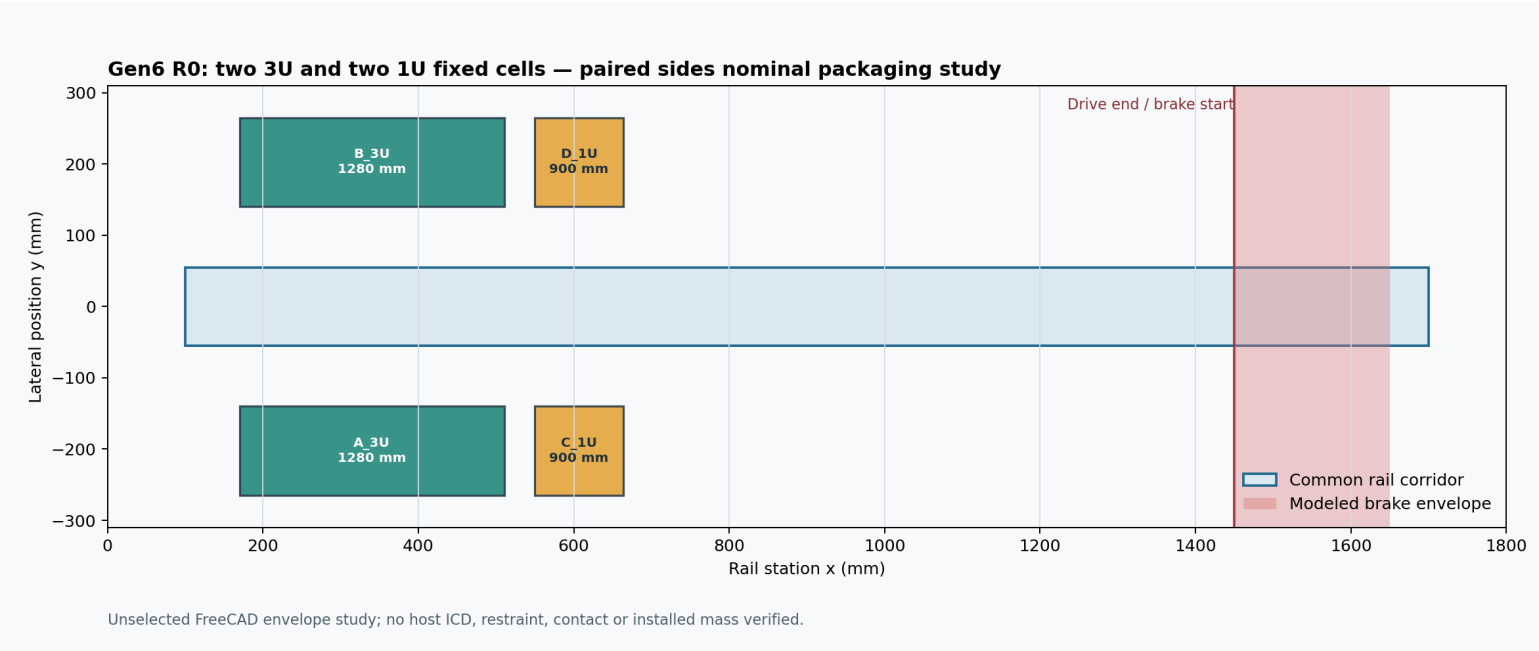
First-article gap

Two STEP envelopes exist; ~9.35 kg remains for all omitted hardware under a surrogate parity screen.

1 km/s remains separate long-range research; no command range or motor rating established.

Paired cells improve the nominal acceleration path

FUTURE WORK



Staggered

Last cell: 330 mm to brake; 5.56 g ideal mean for 6 m/s.

Paired

At least 900 mm per cell; 2.04 g largest ideal mean at 6 m/s.

26 nominal FreeCAD/STEP states pass represented-solid checks; locks, contact, docking and host fit remain open.

Next milestone: close G0 and G1 before any build

FUTURE WORK

Specify	Real customer targets, tolerances and provider interface
Decide	Complete motor, source, brake, contact and installed BOM
Measure	Field, contact, release, arrest, loads and cycle behavior
Review	Drawings, BOM, procedures, uncertainty and review decision

Core literature and standards (1/2)

REFERENCES

- [1] JAXA, JEM Payload Accommodation Handbook, Vol. 8: Small Satellite Deployment ICD.
- [2] Cal Poly, CubeSat Design Specification, Rev. 14.1, 2022.
- [3] C. Foster et al., ‘Constellation phasing with differential drag on Planet Labs satellites,’ J. Spacecraft Rockets 55(2), 2018.
- [4] H. Feng et al., ‘Design and reachable domain analysis of on-orbit electromagnetic launcher for CubeSats,’ Int. J. Aerospace Eng., 2025.
- [5] Y. Zhao et al., ‘Design and analysis of a new deployer for the in-orbit release of multiple stacked CubeSats,’ Remote Sensing 14(17), 4205, 2022.

Methods and project records (2/2)

REFERENCES

• [6] Y. Zhao et al., ‘Simulation analysis and experimental verification of the transport characteristics of a high-volume CubeSat storage device,’ Aerospace 12(6), 466, 2025.

• [7] K. Halbach, ‘Design of permanent multipole magnets with oriented rare earth cobalt material,’ Nucl. Instrum. Methods 169, 1980.

• [8] NASA, General Environmental Verification Standard (GEVS), GSFC-STD-7000A, 2013.

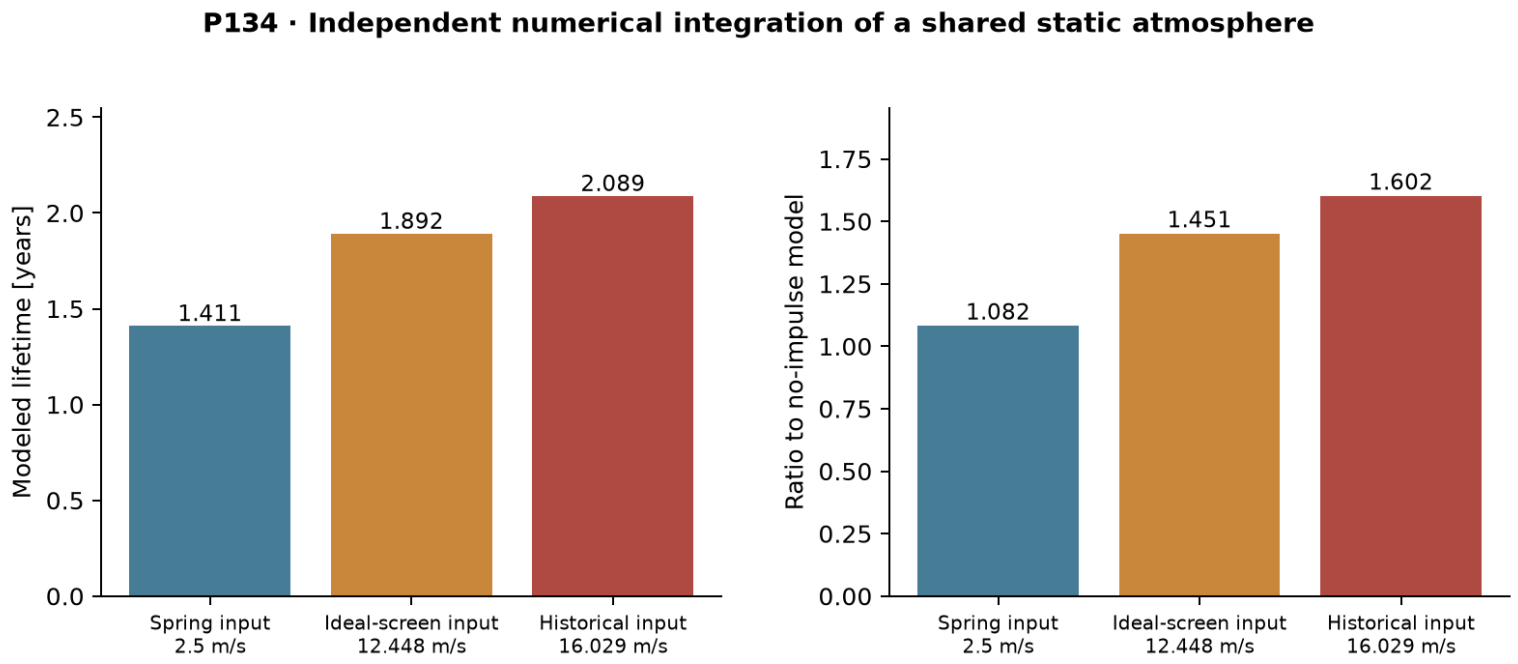
• [9] D. A. Vallado, Fundamentals of Astrodynamics and Applications, 4th ed., 2013.

• [10] A. Mishra, VOLLEY-thesis: manuscript, baseline, provenance and validation records, evidence snapshot 5b92edc, 20 Oct 2026.

• [11] VOLLEY, Gen6 G0 mission and article requirements, 12 Oct 2026 (study requirements).

What the reconciled numerical record establishes

EVIDENCE APPENDIX



450 km start; BC 61 kg/m²; stop at 120 km perigee. Speeds are imposed inertial inputs.
Numerical agreement does not validate atmospheric lifetime or an achieved release speed.

1.602 model ratio

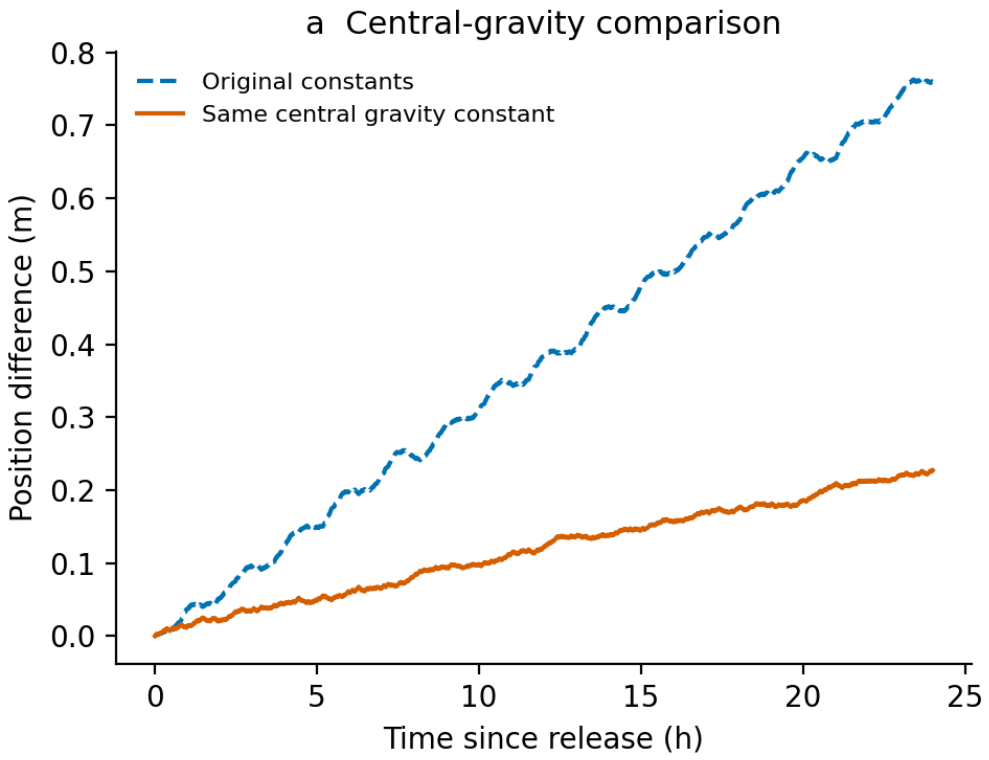
Separate integration, same static atmosphere. Actual lifetime is unvalidated.

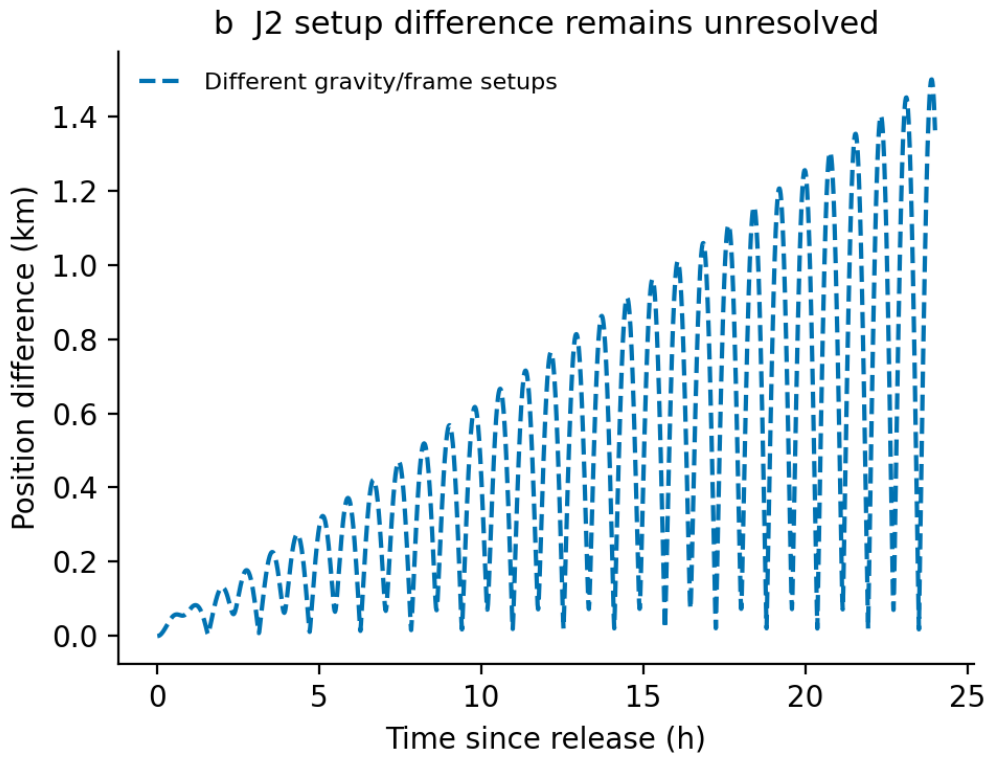
Failures retained

22 Gen5/R1 intersections; 126.562 kg model; archived mission screens fail. New Gen6 targeting is separately bounded.

Native solvers: agreement and unresolved differences

EVIDENCE APPENDIX



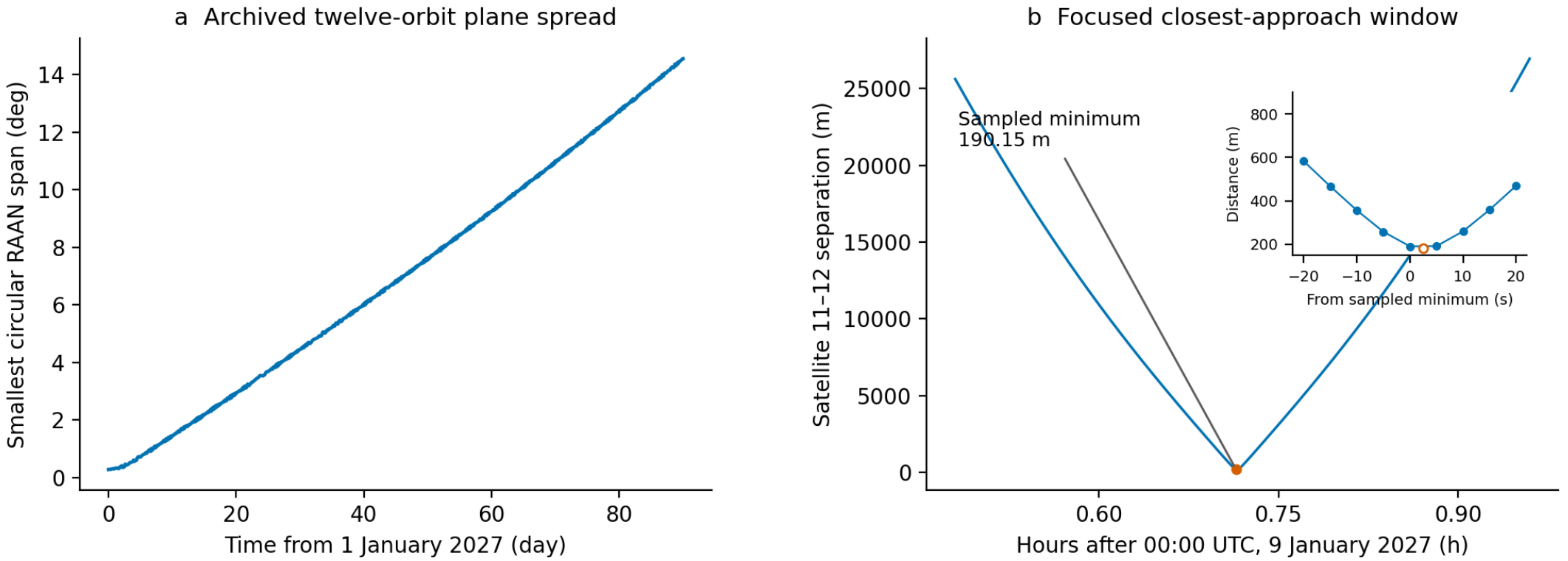


Native GMAT R2026a versus Orekit 13.1.6; identical prescribed initial Cartesian state, 16.029 m/s relative input.
Elapsed output epochs are aligned. The J2 comparison is not a matched-force validation; its residual must not be hidden.

Prescribed states; no release-accuracy or hardware validation. J2 frames are not harmonized.

Archived campaign: sampled proximity has a limit

EVIDENCE APPENDIX

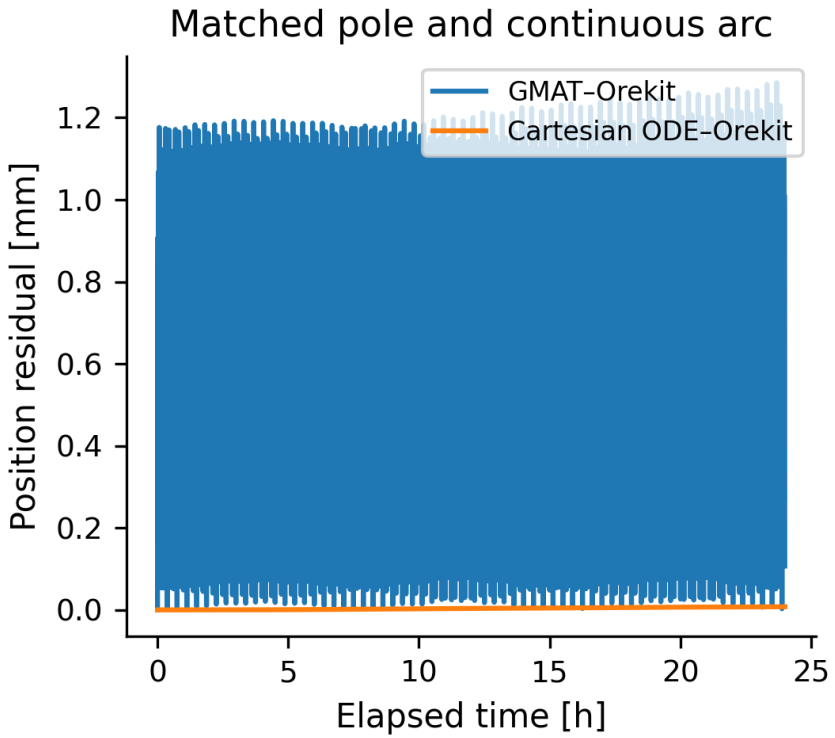


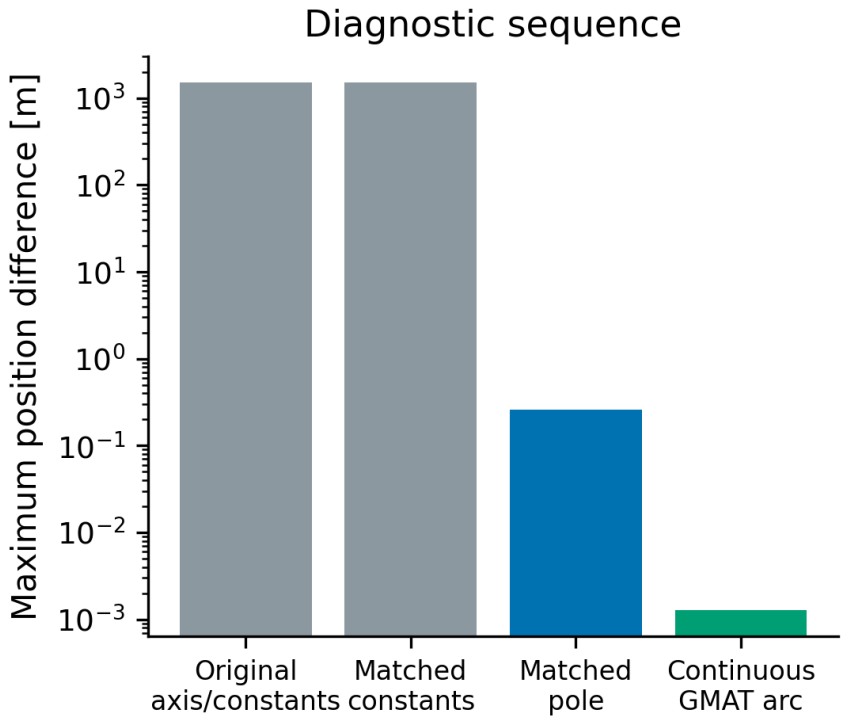
GMAT R2026a archived R1: preloaded post-release orbits, not a coupled host campaign. The left plot is downsampled.
The 190.15 m sampled minimum and 180.29 m linear interpolation do not establish continuous clearance or collision probability.

190.15 m sampled; 180.29 m interpolated. Continuous clearance remains unproven.

The J2 setup discrepancy now has a bounded resolution

EVIDENCE APPENDIX



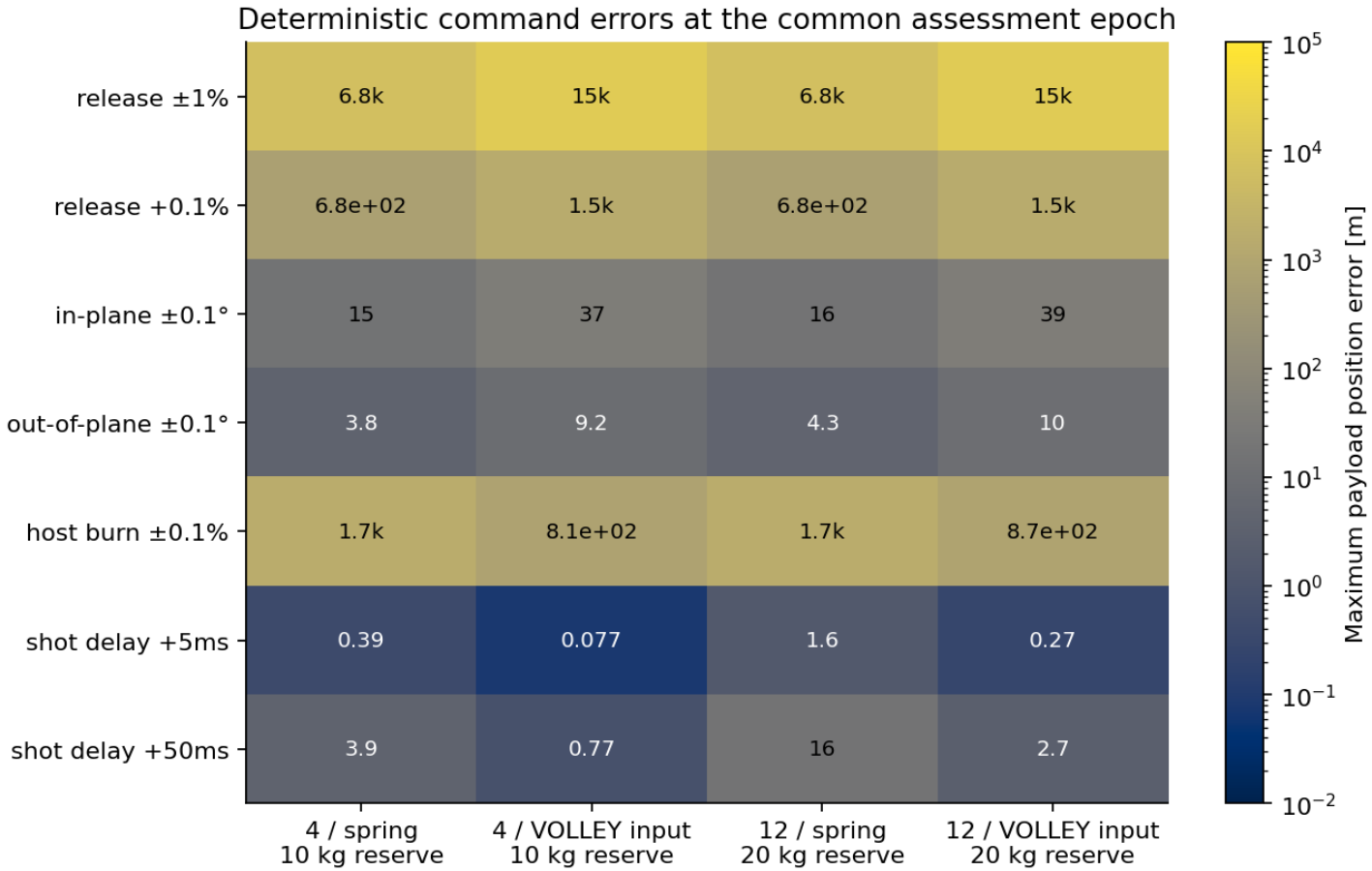


450 km, 51.6°; one day, J2 only. Shared prescribed GMAT EarthFixed pole. First three bars: SciPy versus repeated-stop GMAT; last: Orekit versus continuous GMAT. Different numerical histories are declared; no flight-accuracy or motor validation.

1.285 mm maximum over one day. Shared orientation; no physical orbit or release-accuracy claim.

Precise delivery needs a measured control budget

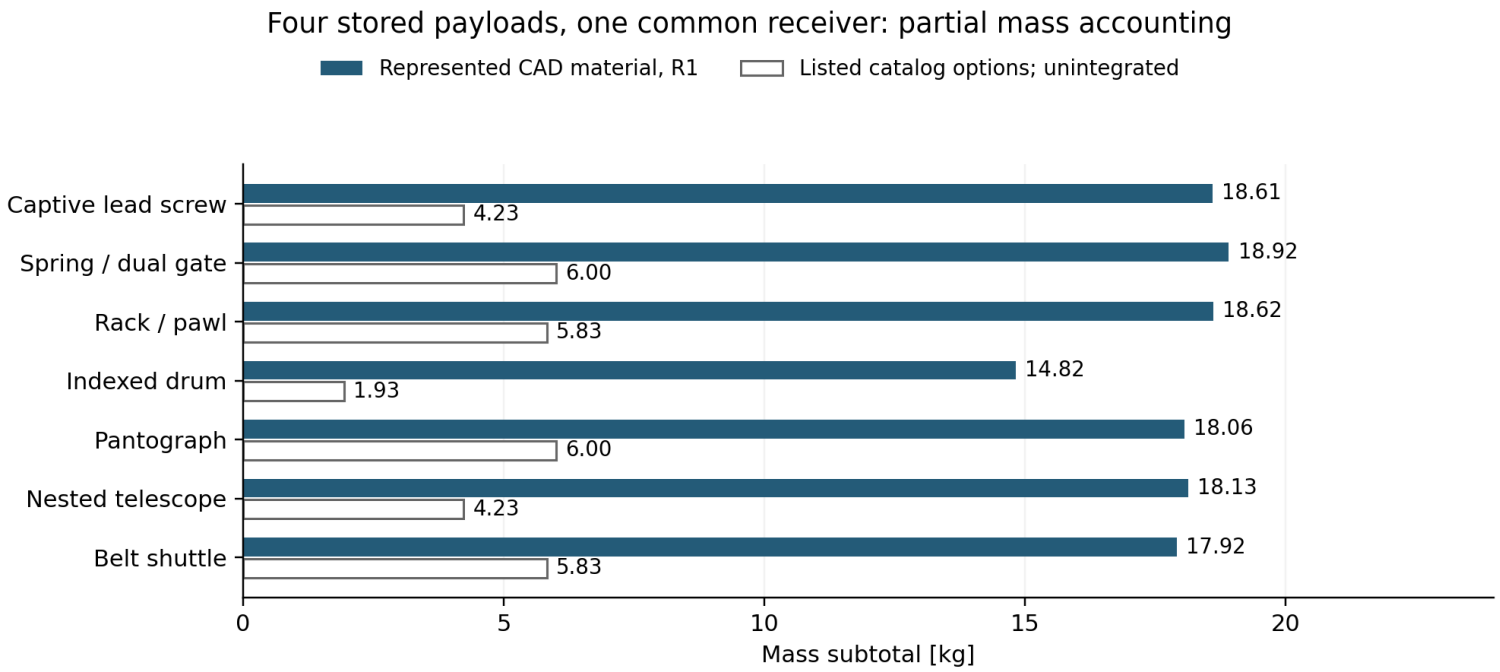
FUTURE-WORK APPENDIX



+0.1% imposed release-speed bias gives ~1.53 km error. Numerical agreement is not flight precision.

Seven feeders on one four-payload comparison boundary

FUTURE-WORK APPENDIX



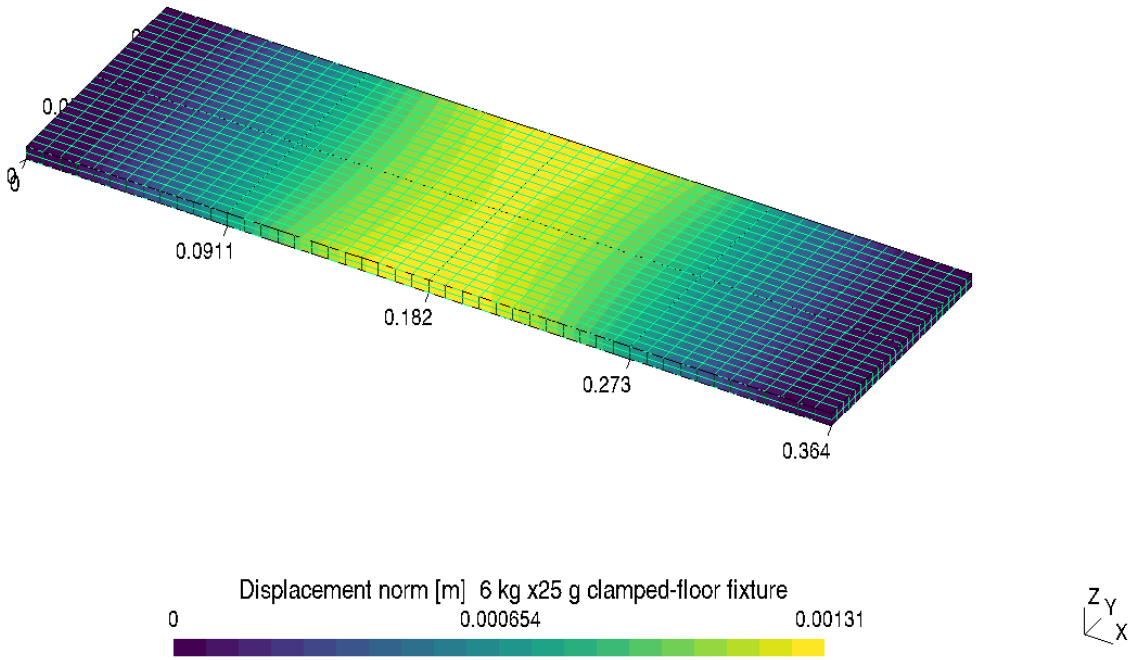
FDR09 / R1 · two 1U + two 3U · complete installed mass, CG and inertia unestablished · no eligible downselect

- Two 1U + two 3U; one shared receiver
- Seven concepts; ten checked native/STEP pairs
- Separate R2 nominal payload sweep: eight collisions to zero
- Full installed mass, loaded support and custody remain unverified
- No eligible architecture selection

CAD material and unintegrated catalog options are partial accounting; no mass ranking or hardware qualification.

Native mesh and result view: keep the fixture explicit

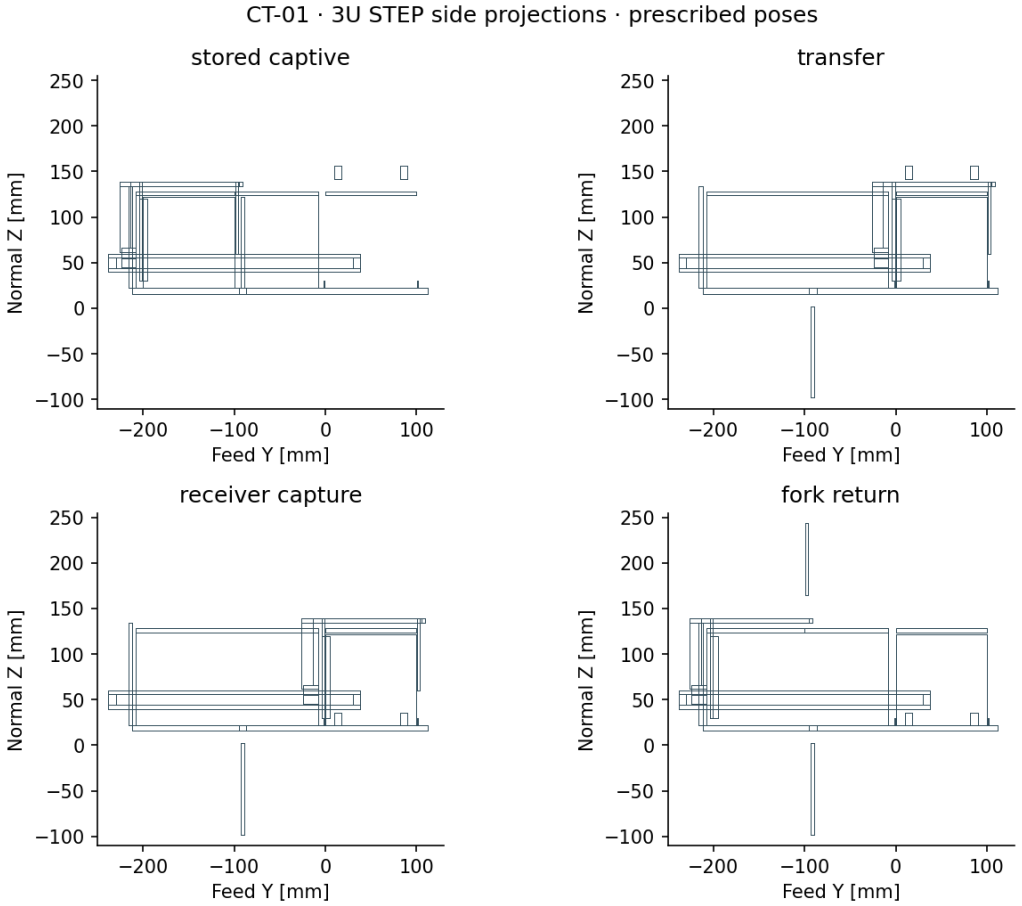
EVIDENCE APPENDIX



Displacement norm in metres. Actual joints absent; peak support stress is not converged or a strength pass.

Continuous feed geometry exposes the remaining safety work

FUTURE-WORK APPENDIX

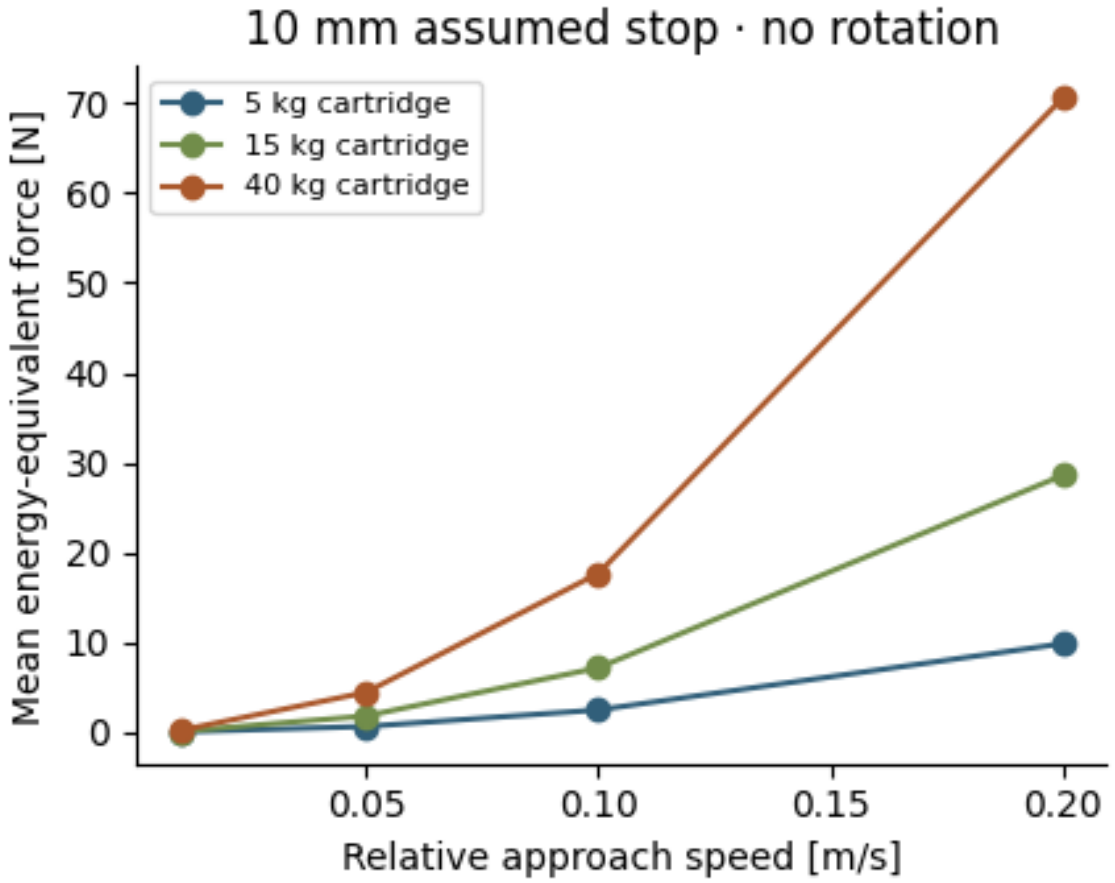


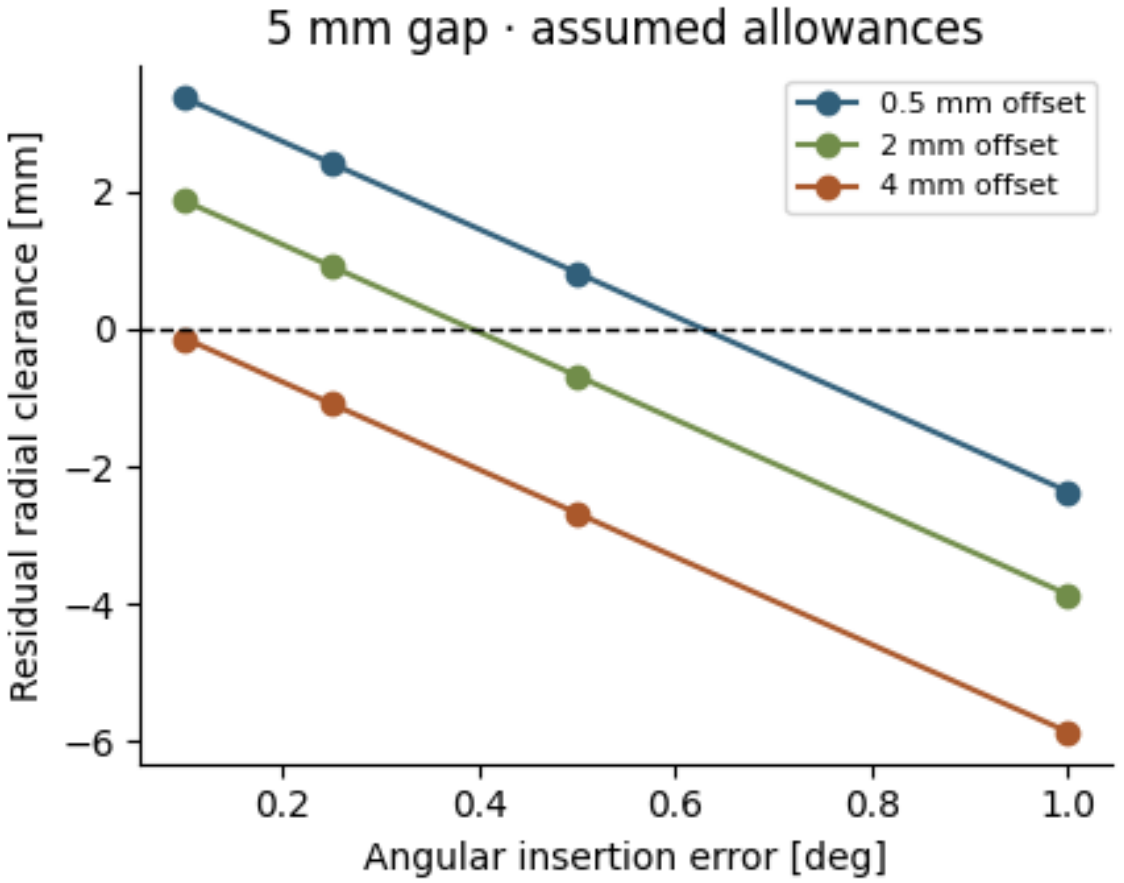
- 20 continuous 1U/3U phase checks
- Downward gate needs under-deck volume
- Stop travel and adverse fit cases fail
- Real latch/custody and drive remain open

Native CAD and nominal translation proof do not establish full mechanism safety or a selected feeder.

Orbital reload: define the interface before promising a service

FUTURE-WORK APPENDIX

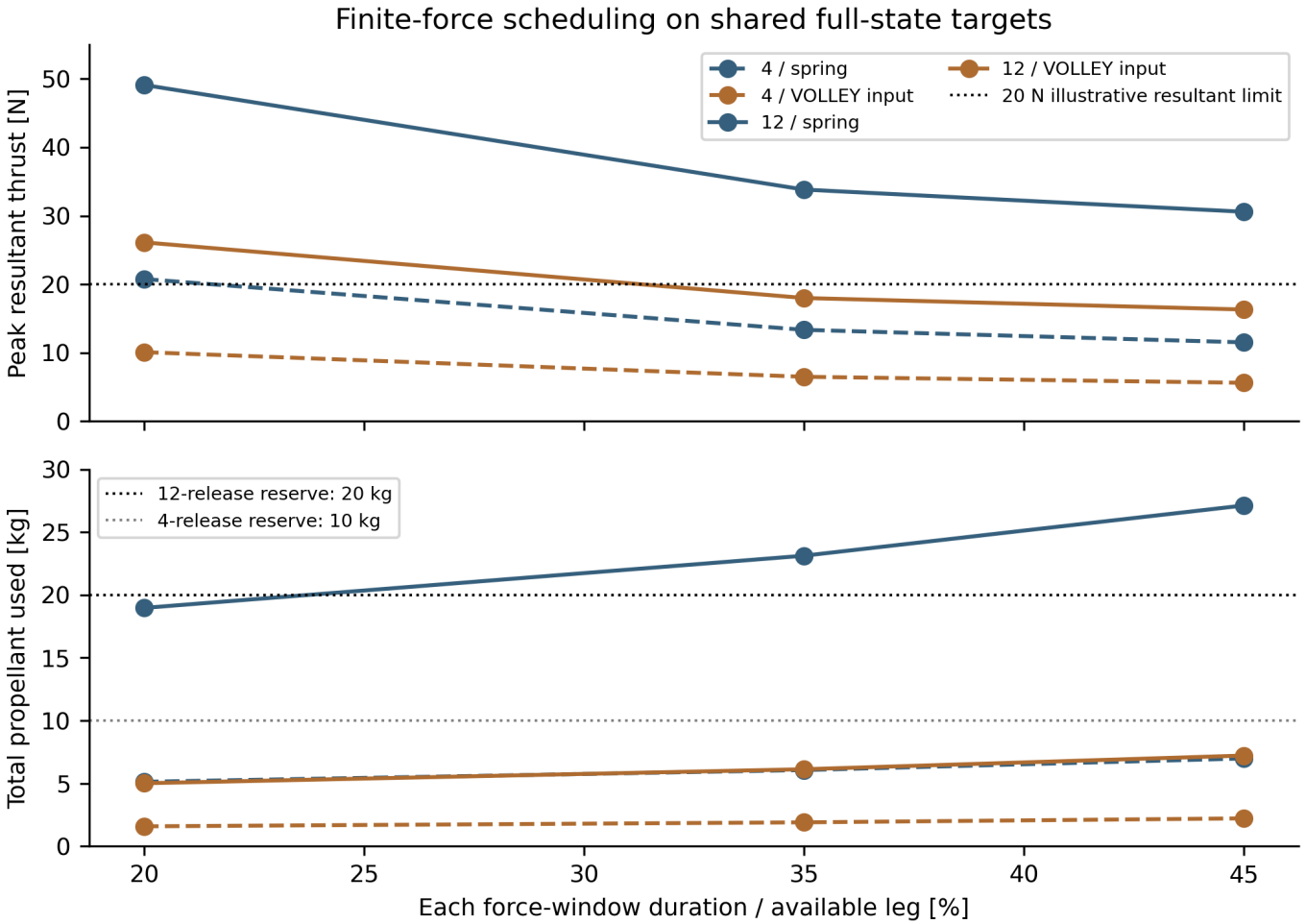




108 capture / 36 fit sensitivities. No integrated reload mechanism; ground loading remains the assumption.

Finite-duration thrust changes the resource comparison

FUTURE-WORK APPENDIX

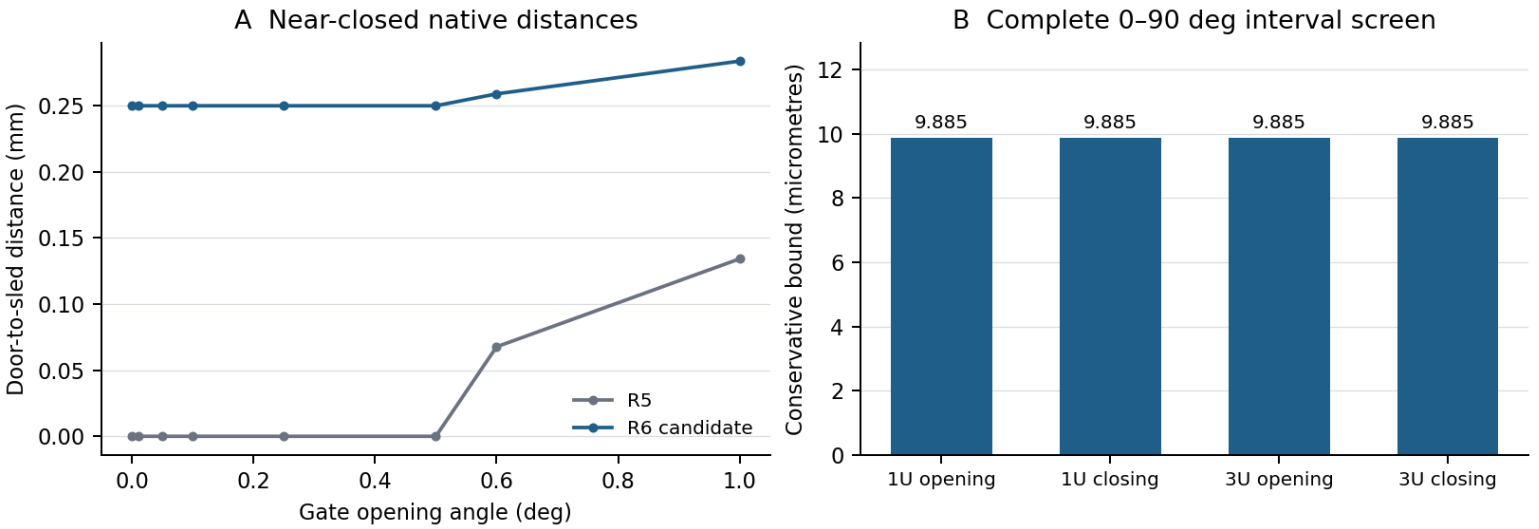


- Four releases fit the 20 N / 10 kg screen
- Twelve spring inputs exceed 20 N / 20 kg
- Both twelve-case policies close with 30 kg / 100 N
- Device mass still exceeds saved propellant
- Nearly 9°/s reversals if one fixed thruster; host allocation unverified

Finite-force translational targets; imposed release speeds. No rated host, motor, flight precision or physical clearance.

Gate relief passes the complete-angle nominal screen

FUTURE-WORK APPENDIX

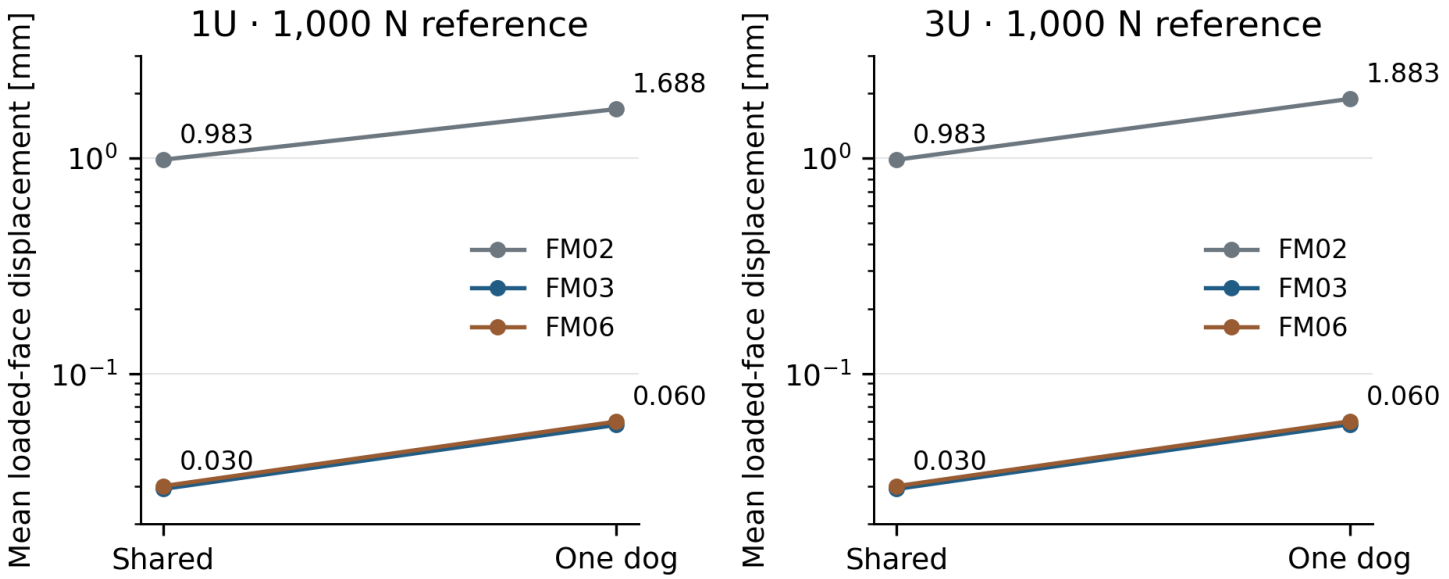


- Panel lower edge relieved by 0.25 mm
- Four full opening/closing intervals checked
- Eight assembly and two gate STEP readbacks
- Nominal numerical bound; loaded fit and actuation remain separate

Smallest guarded bound: 0.0098846 mm. No manufactured-clearance, contact-convergence or qualification claim.

Component stiffness improves within an idealized fixture

FUTURE-WORK APPENDIX

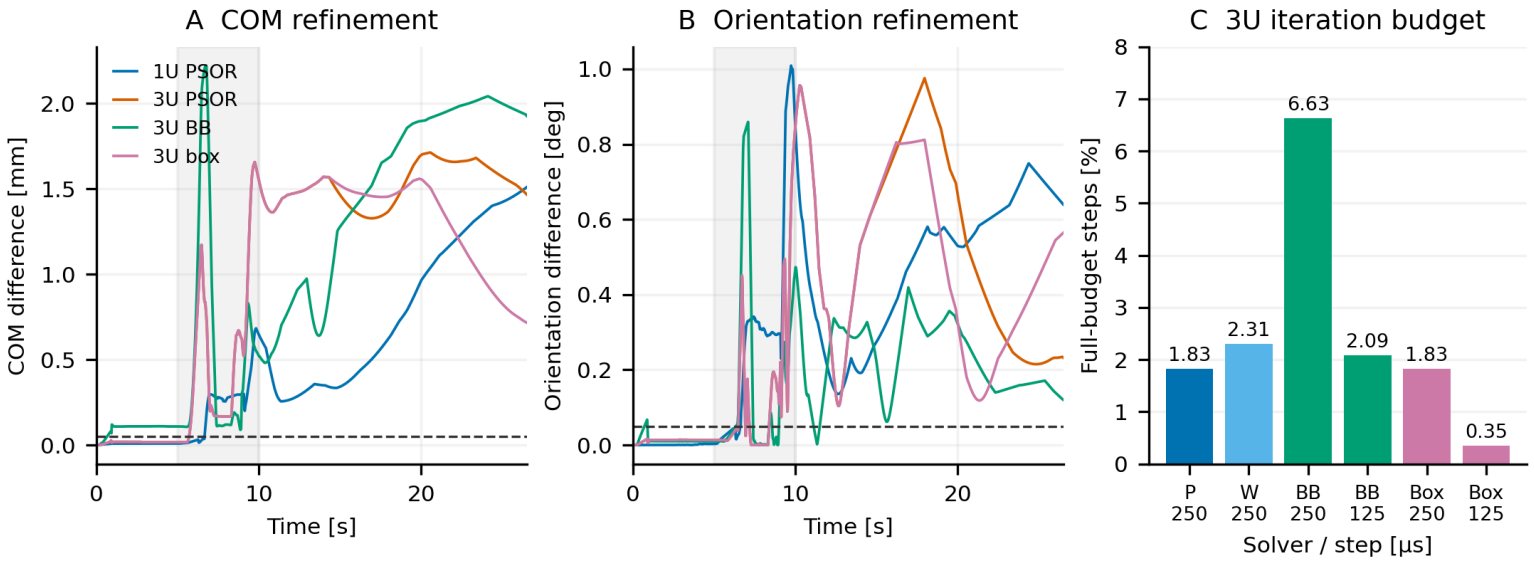


- 12 actual-FM06 plus 40 historical receiver analyses
- 3U one-dog mean: 1.883 to 0.0600 mm
- Clamped bores do not bound assembled stiffness
- 1,117/1,215 adverse guide-fit residuals nonpositive

Assumed elastic properties; no selected material allowable, real joint-contact acceptance or hardware testing.

Contact diagnosis retains the numerical acceptance criteria

FUTURE-WORK APPENDIX



- Instrumented baseline reproduced byte for byte
- PSOR, BB and exact-box refinements exceed pose limits
- Sensitivity develops during feed deceleration
- No qualified correction or contact-force rating

Nominal FM07-R6 rotary screens pass separately; FM06 contact results do not validate that revised assembly.